

Qns	Answer	Marks
9(a)(i)	distance = $0.28 \text{ AU} \times (1.5 \times 10^{11}) = 4.2 \times 10^{10} \text{ m}$	B1
	$I = \frac{P}{4\pi r^2} = \frac{3.8 \times 10^{26}}{4\pi (4.2 \times 10^{10})^2}$	B1
	$I = 1.714 \times 10^4 = 1.7 \times 10^4 \text{ W m}^{-2}$	
9(a)(ii)	$P_{\text{received}} = IA(\text{efficiency})$ $P_{\text{received}} = (1.7 \times 10^4)(6.5)(0.28)$ $P_{\text{received}} = 3.1 \times 10^4 \text{ W}$	C1 A1
9(b)(i)	$P_r = \left(\frac{I}{c} + R \right) = \left(\frac{1.7 \times 10^4}{3.00 \times 10^8} \right) \times \frac{100}{99.8} \times \frac{0.15}{100}$ $= 8.5 \times 10^{-8} \text{ Pa}$	M1
9(b)(ii)	$E_k = \frac{1}{2}mv^2$ $p = mv$ Showing BOTH the above to be awarded 1 mark	M1
	$E_k = \frac{1}{2}m\left(\frac{p}{m}\right)^2$	M1
	Therefore $E_k = \frac{p^2}{2m}$	
9(b)(iii)	$F = PA = (8.5 \times 10^{-8})(6.5) = 5.5 \times 10^{-7} \text{ N}$	C1
	Using $E = \frac{p^2}{2m}, F = \left(\frac{N}{t}\right)\Delta p = \left(\frac{N}{t}\right)\sqrt{2mE}$	C1
	$5.5 \times 10^{-7} = \left(\frac{N}{t}\right)\sqrt{2(4 \times 1.66 \times 10^{-27})(5.0 \times 10^6 \times 1.6 \times 10^{-19})}$ $\frac{N}{t} = 5.4 \times 10^{12} \text{ s}^{-1}$	A1
9(c)	$\text{activity} = \frac{\text{count rate}}{\text{sensitivity}} = \frac{3.6 \times 10^5}{1.2 \times 10^4}$ $= 30 \text{ GBq}$	C1 A1

Qns	Answer	Marks
9(d)(i)	$\lambda_{\alpha} = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{20} = 0.035 \text{ min}^{-1}$ $\lambda_{\beta} = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{10} = 0.069 \text{ min}^{-1}$	A1 A1
9(d)(ii)	$\lambda_{\text{eff}} = 0.01\lambda_{\alpha} + 0.99\lambda_{\alpha}$ $= 0.01(0.035) + 0.99(0.069)$ $= 6.87 \times 10^{-2} \text{ min}^{-1}$	M1
9(d)(iii)	$N_0 = \frac{2.2 \times 10^{-11}}{13(1.66 \times 10^{-27})} = 1.0 \times 10^{15}$ $N = N_0 e^{-\lambda t}$ $= (1.0 \times 10^{15}) e^{-(6.9 \times 10^{-2})(15)}$ $= 3.6 \times 10^{14}$	C1 C1 A1
9(d)(iv)	${}^{13}_7\text{N} \rightarrow {}^{13}_6\text{C} + {}^0_1\text{e} + \text{neutrino}$ (accept anti-neutrino and symbols $\nu, \bar{\nu}$)	A1
9(e)	Any of the following: relate distance from the Sun to intensity via the inverse-square law solar activity (e.g., solar flares) panel orientation, dust accumulation temperature effects on solar cell efficiency	B1